NHL Play by Play Featuring Detroit 2023-24

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Installing Data

```
#install.packages("hockeyR")
#install.packages("devtools")
devtools::install_github("danmorse314/hockeyR")
devtools::install_github("jamesmartherus/shotplot")
devtools::source_url("https://raw.githubusercontent.com/mrbilltran/the-win-column/master/nhl_rink_
library(hockeyR)
library(sportyR)
library(tidyverse)
library(devtools)
library(png)
library(grid)
library(jsonlite)
library(curl)
library(shotplot)
library(knitr)
geom_shotplot <- function(...){
  list(annotation_custom(grid::rasterGrob(png::readPNG(here::here("man/figures/full-rink.png")),
                                                                  width = unit(1,"npc"),
                                                                  height = unit(1,"npc"))),
  geom_point(),
  coord_flip())
}
```

Introduction:

Being a huge Detroit Red Wings fan like myself, I decided to dive into a data set where I could analyse some of the pro's and the cons that the Wing's faced this season. Unfortunately the Wing's did not make the Stanley Cup Playoff's by a very close call; they lost the head to head match over the Capitals and just fell short as they were tied for points. In the National Hockey League a win counts as 2 points a tie counts as 1 and a loss counts as 0, so every game is critical. Being a hockey player I decided to do a small scouting report on the Wings analyzing four main things: penalties taken, blocked shots, power play conversions, and goals scored. For these categories what can the Wing's do better to be prepared for the 2024-25 season in order to make the 2025 Stanley Cup Playoff's?

Data Sets Used:

The data set that I chose to use is a set that has every single play from the NHL from the start to the end of the 2023-2024 regular season. The 'pbp' data set had originally 425,751 observations and 138 variables. When I filtered it just to the Red Wings pbp to be 'Wingspbp' it had 10094 observations and 138 variables. Then I filtered 'Wingspbp' to just Red Wings goals to 'Wingsgoals' which had 280 observations and 138 variables. Each of these three data sets were used a lot throughout my whole project.

```
pbp <- load_pbp('2023-24')##Every NHL team play by play
Wingspbp <- pbp|> filter(event_team == "Detroit Red Wings") ##Redwings play by play
Wingsgoals <- Wingspbp|> filter(event_type == "GOAL")|> ##Redwings goals
  mutate(secondary_type = fct_collapse(secondary_type,
                                       wrist_snap = c("wrist", "snap"),
                                       deflect_tip = c("deflected", "tip-in"),
                                       trick = c("between-legs", "poke", "wrap-around")))

head(pbp)|> kable()
```

event_type	event	secondary_type	event_team	event_team_type	description	period	period_seconds	perio
PERIOD_START	period-start	NA	NA	NA	Start of Period 1	1	0	
FACEOFF	faceoff	NA	Pittsburgh Penguins	home	Sidney Crosby faceoff won against Connor Bedard	1	0	
HIT	hit	NA	Pittsburgh Penguins	home	Jake Guentzel hit Seth Jones	1	35	
SHOT	shot-on-goal	backhand	Pittsburgh Penguins	home	Bryan Rust shot on goal saved by Petr Mrazek	1	40	
SHOT	shot-on-goal	wrist	Chicago Blackhawks	away	Kevin Korchinski shot on goal saved by Tristan Jarry	1	59	

event_type	event	secondary_type	event_team	event_team_type	description	period	period_seconds	perio
STOP	stoppage	NA	NA	NA	Stoppage in play (puck-in-netting)	1	62	

```
head(Wingspbp)|> kable()
```

event_type	event	secondary_type	event_team	event_team_type	description	period	period_seconds	perio
FACEOFF	faceoff	NA	Detroit Red Wings	away	Dylan Larkin faceoff won against Nico Hischier	1	0	
SHOT	shot-on-goal	slap	Detroit Red Wings	away	David Perron shot on goal saved by Vitek Vanecek	1	8	
HIT	hit	NA	Detroit Red Wings	away	David Perron hit Kevin Bahl	1	12	
MISSED_SHOT	missed-shot	snap	Detroit Red Wings	away	Alex DeBrincat shot missed wide of net	1	20	
FACEOFF	faceoff	NA	Detroit Red Wings	away	J.T. Compher faceoff won against Nico Hischier	1	28	
BLOCKED_SHOT	blocked-shot	NA	Detroit Red Wings	away	Moritz Seider shot blocked by Dawson Mercer	1	31	

```
head(Wingsgoals)|> kable()
```

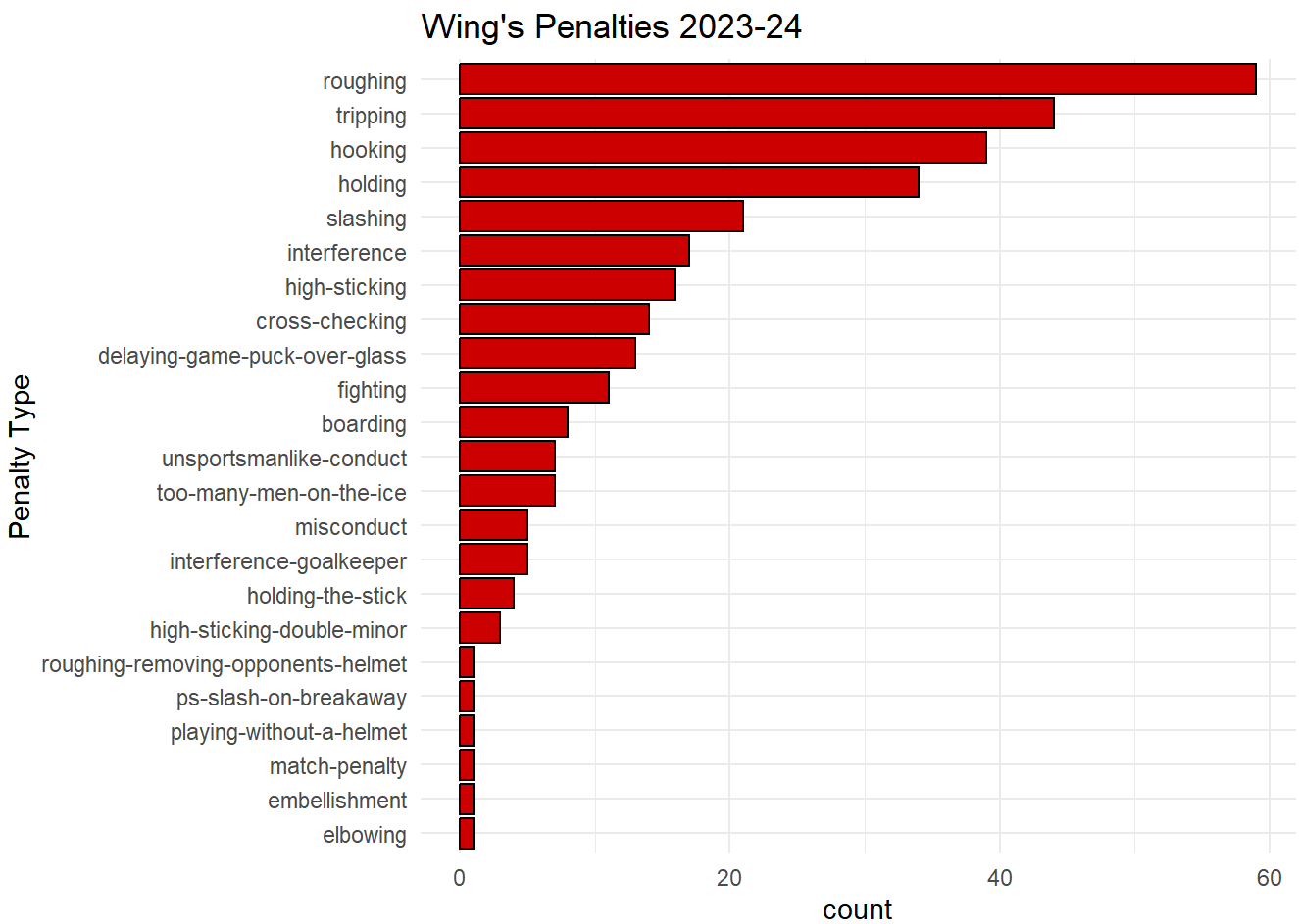
event_type	event	secondary_type	event_team	event_team_type	description	period	period_seconds	period_sec
GOAL	goal	deflect_tip	Detroit Red Wings	away	Daniel Sprong deflected, unassisted	2	734	
GOAL	goal	wrist_snap	Detroit Red Wings	away	Alex DeBrincat wrist, assists: Dylan Larkin, Shayne Gostisbehere	3	643	
GOAL	goal	wrist_snap	Detroit Red Wings	away	Robby Fabbri wrist, assists: J.T. Compher, Jake Walman	3	1166	
GOAL	goal	trick	Detroit Red Wings	home	Daniel Sprong poke, assists: Alex DeBrincat	1	311	
GOAL	goal	wrist_snap	Detroit Red Wings	home	Alex DeBrincat snap, assists: Shayne Gostisbehere	1	875	
GOAL	goal	slap	Detroit Red Wings	home	Lucas Raymond slap, unassisted	2	121	

The different types of penalties the Red Wings took and the amount per penalty:

```
Wings_penalty <- Wingspbp |> filter(event_type == "PENALTY")|>
  group_by(secondary_type)|>
  summarise(count = n())|>
  mutate(secondary_type = fct_reorder(secondary_type, count))


ggplot(data = Wings_penalty, aes(x = secondary_type, y = count))+
  geom_col(fill = "red3", colour = "black")+
  coord_flip()+
```

```
  theme_minimal()+
  labs(x = "Penalty Type", title = "Wing's Penalties 2023-24")
```



Looking at the bar plot we can see that the Wings had a big problem with roughing penalties this season. A roughing penalty is defined as “using arms, hands, or stick in a careless or reckless manner”, to me roughing is a discipline issue with the team that needs to be assessed by the coaching staff. All though call’s by the ref can go both ways it is up to the player to stay calm and think about the team first. A penalty like Fighting is more disciplined because it usually doesn’t result in the opposing team getting a power play (odd man advantage for 2 minutes) instead both players go to the penalty box. In order to make it to playoffs next season the Wing’s need to limit the penalties out of frustration, penalties will happen but being more disciplined as a team would be the answer to this question.

The proportion of penalties the Red Wings took per game compared to other NHL teams:

```
pbp|> filter(event_type == "PENALTY")|>
  group_by(event_team)|>
  summarise(pen_per_game = n()/82)|> ##82 games in the season
  mutate(event_team = fct_reorder(event_team, pen_per_game),
         pen_per_game = round(pen_per_game, 2))|>
```

```r
arrange(pen_per_game)|>
kable(caption = "NHL Average Penalties per Game", row.names = TRUE, align = 'c')
```

NHL Average Penalties per Game

	event_team	pen_per_game
1	St. Louis Blues	3.22
2	Vegas Golden Knights	3.24
3	Seattle Kraken	3.27
4	Pittsburgh Penguins	3.35
5	New York Islanders	3.52
6	Chicago Blackhawks	3.56
7	Columbus Blue Jackets	3.57
8	Washington Capitals	3.57
9	Winnipeg Jets	3.61
10	Dallas Stars	3.70
11	New Jersey Devils	3.70
12	Buffalo Sabres	3.77
13	Detroit Red Wings	3.82
14	San Jose Sharks	3.82
15	Calgary Flames	3.89
16	Philadelphia Flyers	3.91
17	Los Angeles Kings	4.04
18	Nashville Predators	4.04
19	Ottawa Senators	4.16
20	Carolina Hurricanes	4.23
21	Tampa Bay Lightning	4.23
22	Toronto Maple Leafs	4.24
23	Minnesota Wild	4.28
24	Montreal Canadiens	4.28
25	Colorado Avalanche	4.29
26	Arizona Coyotes	4.39
27	Vancouver Canucks	4.52
28	New York Rangers	4.56
29	Boston Bruins	4.87
30	Edmonton Oilers	5.12
31	Anaheim Ducks	5.38

	event_team	pen_per_game
32	Florida Panthers	6.34

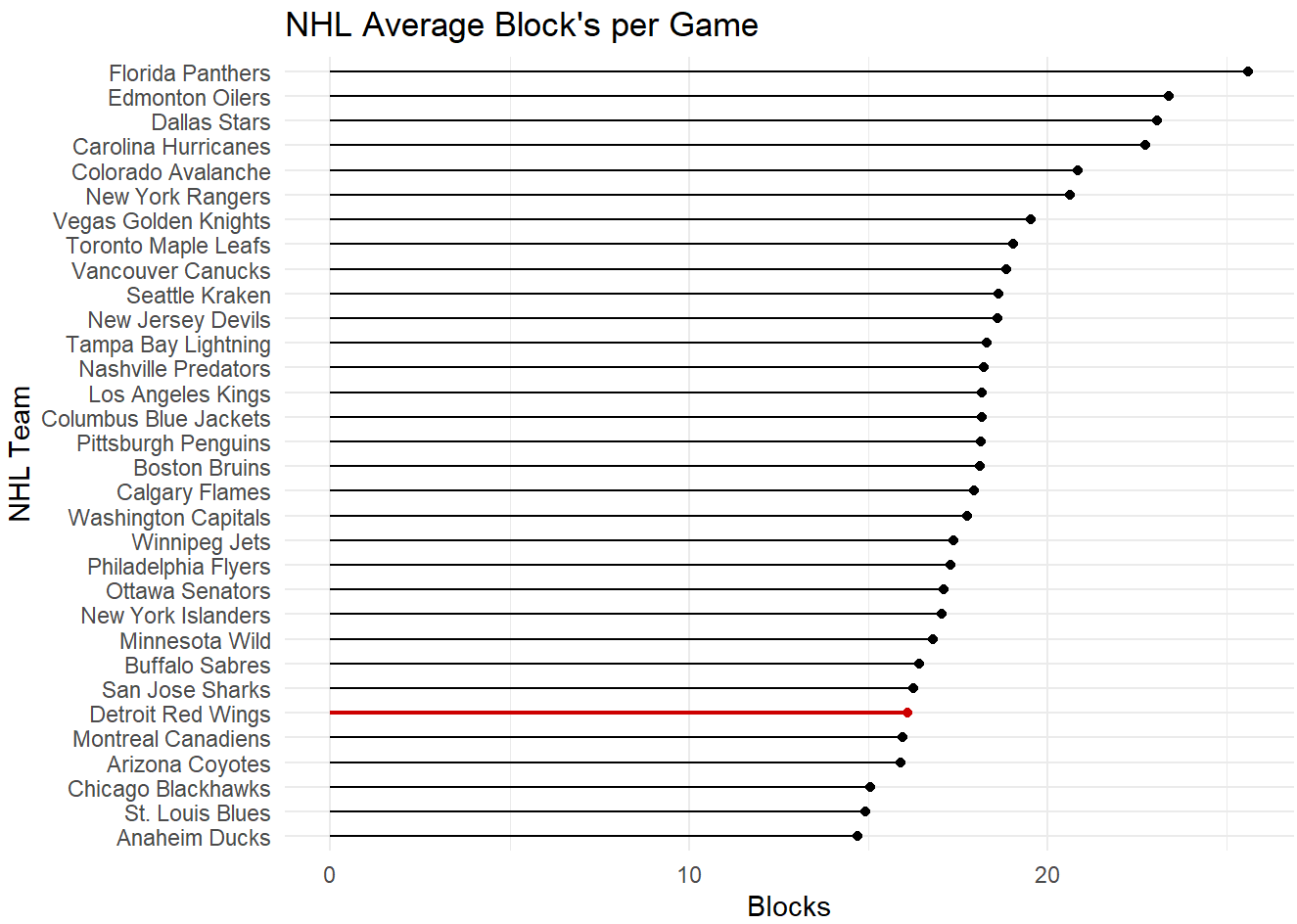
The Detroit Red Wings ranked 13th with an average of 3.82 penalties every game, not terribly bad in terms of discipline but would like to see them bring the average down to closer to 3 penalties a game next season. As you can see teams like Vegas, Dallas, and Islanders are at the top and these are teams that made it into the 2024 Stanley Cup Playoffs. As I said above undisciplined penalties like roughing, tripping, hooking, and slashing need to be limited if the Wing's want a shot at the playoff's next season.

Proportion of blocked shots Red Wings had per game compared to other NHL teams:

```
Wings_blocks <- Wingspbp|> filter(event_type == "BLOCKED_SHOT")|>
  group_by(event_team)|>
  summarise(blocked_per_game = n()/82) ##82 games in the season

nhl_blocks <- pbp|> filter(event_type == "BLOCKED_SHOT")|>
  group_by(event_team)|>
  summarise(blocked_per_game = n()/82)|> ##82 games in the season
  mutate(event_team = fct_reorder(event_team, blocked_per_game))|>
  arrange(desc(blocked_per_game))|>
  filter(!is.na(event_team))

ggplot(data = nhl_blocks, aes(x = event_team, y = blocked_per_game)) +
  geom_segment(aes(xend = event_team, y = 0, yend = blocked_per_game))+
  geom_point() +
  geom_segment(data = Wings_blocks, aes(xend = event_team, y = 0, yend = blocked_per_game),
               color = "red3", linewidth = .8) +
  geom_point(data = Wings_blocks, color = "red3") +
  coord_flip() +
  theme_minimal()+
  labs(x = "NHL Team", y = "Blocks", title = "NHL Average Block's per Game")
```



The Detroit Red Wings ranked 27th in the NHL with an average of 16.1 blocked shots per game which is not good. The Wing's getting the 16.1 up to a 17.5/18 blocks per game would give the Red Wings a better shot to making playoffs next year. This could start by bringing in a guy who is known for his defense and penalty kill skills.

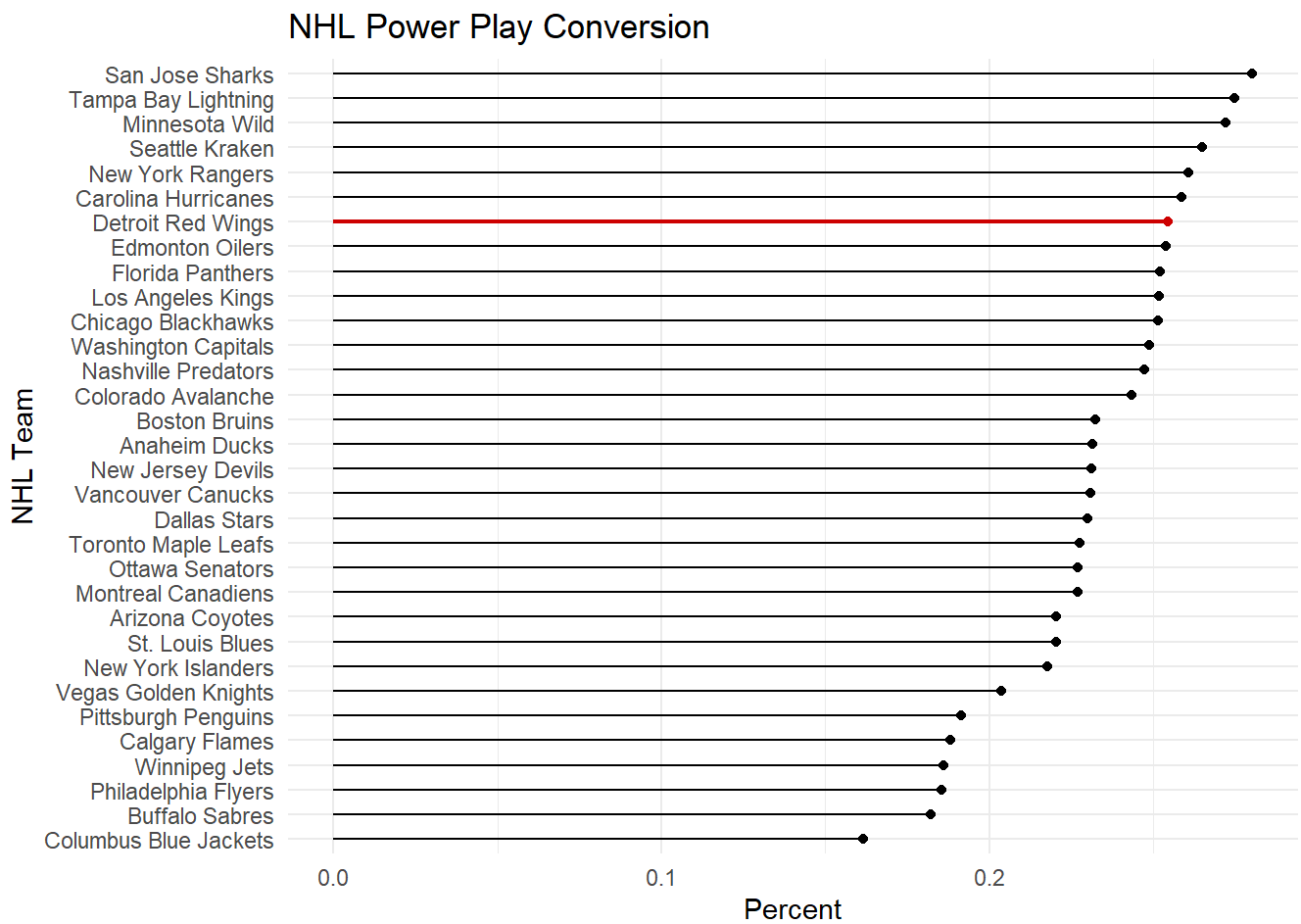
Red Wing's power play percentage compared to other teams:

```
Wings_pp <- Wingsgoals|> mutate(pp_goal = str_detect(strength_code, pattern = "PP"))|>
  select(pp_goal, everything())|>
  filter(!is.na(pp_goal))|>
  group_by(event_team)|>
  summarise(pp_percent = mean(pp_goal))

nhl_pp <- pbp|> filter(event_type == "GOAL")|>
  mutate(pp_goal = str_detect(strength_code, pattern = "PP"))|>
  filter(!is.na(pp_goal))|>
  group_by(event_team)|>
  summarise(pp_percent = mean(pp_goal))|>
  mutate(event_team = fct_reorder(event_team, pp_percent))|>
  arrange(desc(pp_percent))

ggplot(data = nhl_pp, aes(x = event_team, y = pp_percent)) +
  geom_segment(aes(xend = event_team, y = 0, yend = pp_percent)) +
```

```
  geom_point() +
  geom_segment(data = Wings_pp, aes(xend = event_team, y = 0, yend = pp_percent),
               color = "red3", linewidth = .8)+
  geom_point(data = Wings_pp, color = "red3")+
  coord_flip() +
  theme_minimal()+
  labs(x = "NHL Team", y = "Percent", title = "NHL Power Play Conversion")
```



The Detroit Red Wings ranked 9th in the NHL with power their play at 25.5 percent, meaning 25.5 percent of the goals that they scored in the 2023-24 season were on the power play. There is always room for improvement but this is a major improvement for the Wing's and a positive to look at. If the Wing's continue to convert on power play opportunities they will have a good chance of making it to the playoff's next season.

Red Wings goal's scored ranked in 2023-2024 with a shot plot for type of goal scored:

```
pbp|> filter(event_type == "GOAL")|>
  group_by(event_team)|>
  summarise(goals = n())|>
  arrange(desc(goals))|>
  kable(cation = "Total Goal's Scored per Team", row.names = TRUE, align = 'c')
```

	event_team	goals
1	Edmonton Oilers	384
2	Dallas Stars	354
3	Colorado Avalanche	347
4	Florida Panthers	345
5	New York Rangers	334
6	Carolina Hurricanes	320
7	Toronto Maple Leafs	319
8	Vancouver Canucks	312
9	Tampa Bay Lightning	307
10	Boston Bruins	303
11	Vegas Golden Knights	286
12	Nashville Predators	284
13	Detroit Red Wings	280
14	Los Angeles Kings	277
15	Winnipeg Jets	274
16	New Jersey Devils	264
17	New York Islanders	262
18	Pittsburgh Penguins	260
19	Arizona Coyotes	257
20	Calgary Flames	256
21	Ottawa Senators	256
22	Minnesota Wild	254
23	Buffalo Sabres	249
24	Montreal Canadiens	245
25	St. Louis Blues	243
26	Columbus Blue Jackets	239
27	Philadelphia Flyers	237
28	Washington Capitals	231
29	Seattle Kraken	222
30	Anaheim Ducks	205
31	San Jose Sharks	185
32	Chicago Blackhawks	181

```
ggplot(data = Wingsgoals, aes(x = x_fixed, y = y_fixed, color = secondary_type))+
  geom_shotplot()
```

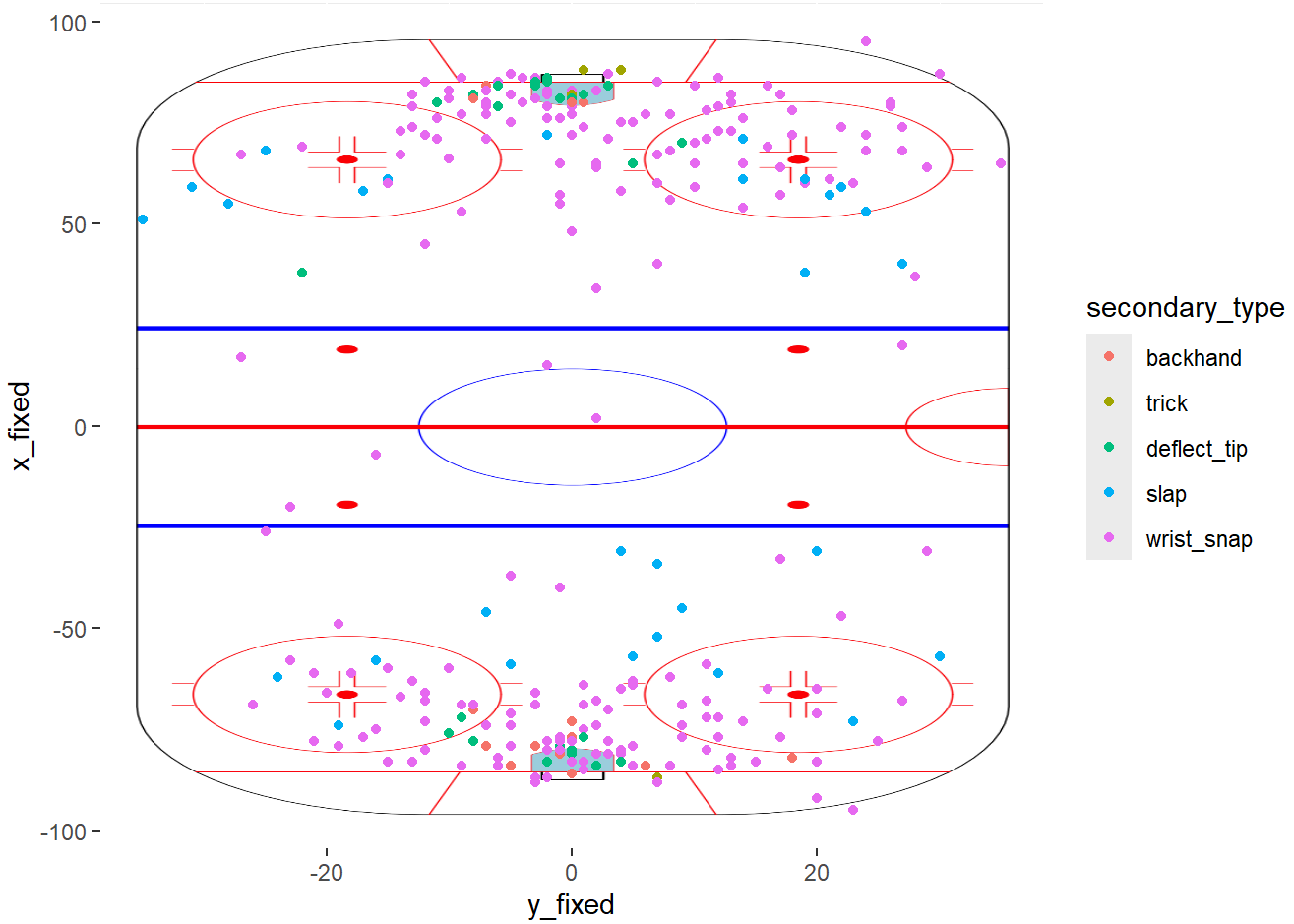


Table: The Detroit Red Wing's ranked 13th in the NHL which is not bad for goals scored this season! If the Wing's continue to produce goal's they should find themselves in a good spot for next year in the rank's for playoffs.

Shot Plot: We can see that most of the goals that were scored were either a wrist shot or snapshot, which mean's they were taken on the forehand of their stick which is common for a goal. The negative end of the rink on the x-fixed from the red line down meant it was an away goal scored and the positive side of the x-fixed meant it was a home goal scored. Each dot represents the area on the rink where the player had shot the puck to score, some are from center ice meaning it was most likely an empty net goal.

Conclusion:

Overall, the Detroit Red Wing's had a good season in terms of offensive production. Some things that the Wing's need to work on is the discipline in their system, this can be addressed by leaders and coaches as we see it can cost them. Only 16 out of the 32 NHL teams make the Stanley Cup Playoff's each year and it is important that each game is taken serious out of the 82 regular season games they play. To wrap up, penalties need to be controlled more, there need to be more blocked shots next year, the Wing's need to

continue to convert on the power play chances, and the Wing’s need to produce more goals in order to make it into the Stanley Cup Playoff’s for next year.